GURWINDERJIT (GARRY BAJWA) SINGH

MANUFACTURING OPERATIONS | AUTOMATION | MAINTENANCE & RELIABILITY

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PROFESSIONAL PROFILE

Manufacturing operations and industrial automation professional with progressive experience in plant and maintenance leadership, electromechanical troubleshooting, CNC equipment, PLC-based machine improvements, and custom production systems. Combines hands-on technical execution with supervisor coordination, preventive maintenance, workforce planning, vendor management, and shop-floor deployment. Electronics and Communication Engineering background; continuing industrial controls training at BCIT.

CORE CAPABILITIES

Operations & Leadership	Production coordination, supervisors and technical teams, capacity planning, training, process improvement, contractor oversight
Maintenance & Reliability	Preventive/corrective maintenance, root-cause analysis, breakdown response, CNC, conveyors, compressors, dust collection, material handling
Controls & Systems	Siemens S7-1200, PLC machine sequencing, sensors, VFDs, motor control, industrial wiring, HMI concepts, networked machinery
Digital Manufacturing	Python, Flask, Socket.IO, barcode traceability, Excel/Pandas/OpenPyXL, KPI dashboards, CNC workflow integration

PROFESSIONAL EXPERIENCE

VAN GOGH DESIGNS **Surrey, BC | Jan 2024 – Aug 2026**

Systems, Automation & Maintenance Manager (Apr–Aug 2026) | Plant & Maintenance Manager (Jan 2025–Mar 2026) | Maintenance Supervisor (Jan–Dec 2024)

- Progressed through maintenance and plant leadership roles, coordinating operations and reliability across CNC machining, frames, upholstery, sewing, shipping, and supporting production equipment.
- Led eight direct reports (four supervisors, two engineers, two machine operators) and supported staffing, task allocation, cross-training, production priorities, and issue escalation across the wider workforce.
- Planned preventive and corrective maintenance; personally diagnosed failures involving CNC routers, spindle drives, sensors, material-handling systems, compressors, forklifts, and production machinery; coordinated specialist contractors when needed.
- Programmed, wired, tested, and commissioned a Siemens S7-1200 PLC retrofit for a fiber-filling machine, replacing manual relay sequencing with timed startup/shutdown logic while retaining contactor and VFD functions.
- Independently developed and deployed a plant-wide barcode production-traceability platform using Python, Flask and Socket.IO, supporting live departmental progress, time-based work records, labour analysis, exception handling, and operational reporting.
- Connected shop-floor production data with custom CNC-side software, G-code validation, and leftover-material tracking workflows; developed additional CAD/CAM and label-processing utilities to reduce manual production administration.
- Coordinated equipment repairs, technical upgrades, vendor work, supervisor training, and daily production adjustments to support safe operations and manufacturing continuity.

Electronics Engineer / Engineering Team Lead **Dec 2022 – Jan 2024**

Tinari Energy Management Services Inc. | Canada

- Worked with a multidisciplinary development team on a large-scale cable-driven concrete 3D-printing prototype, integrating electrical/electronic hardware, motor drives, motion-control subsystems, sensors, and operator interfaces.
- Led hands-on controls/electronics integration, wiring, firmware-related development, troubleshooting, prototype testing and physical printing trials; collaborated with mechanical and engineering team members.

PROFESSIONAL EXPERIENCE (CONTINUED)

Production Supervisor / Technical Manufacturing Support **Oct 2021 – Nov 2022**
Signhub Canada | Surrey, BC

- Supervised daily custom-manufacturing work across signage, CNC cutting, laser/printing equipment, fabrication and installation support; coordinated staff assignments, materials, deadlines, quality and production issues.
- Diagnosed machine and workflow problems, translated designs into shop-floor instructions and supported CAD/CAM preparation, production sequencing and equipment operation.

SMT Operator (Part-Time) **Feb 2021 – Aug 2021**
Motorola Solutions / Avigilon | Canada

- Supported Juki pick-and-place SMT production, printed circuit board assembly, inspection, and electronics manufacturing/testing workflows.

Electronics / Electrical Engineer **Feb 2019 – Nov 2020**
Bajwa Smart Tech

- Developed and troubleshot embedded-control hardware and firmware involving microcontrollers, motor control, sensors, circuit diagrams, PCB systems, and prototype assembly/testing.

SELECTED ENGINEERING RESULTS

- **Plant-wide production traceability:** Took a departmental scan-tracking idea through software design, operational rollout, employee training, dashboard/reporting functions, and expansion to connected manufacturing applications.
- **PLC machine modernization:** Replaced a manual multi-step relay-control sequence with Siemens S7-1200 programmed timing and simplified operator Start/Stop controls.
- **CNC equipment recovery:** Investigated spindle overload, isolated a damaged spindle power cable, and validated production cutting after repair; also supported tool-changer and conveyor-alignment recovery.
- **Forklift workstation reliability:** Designed fused DC-DC power distribution for a printer and tablet, replacing UPS units that ran about two hours during an eight-hour shift; documented approximately $110 in replacement components per setup versus $2,400 in UPS hardware for the prior two-unit arrangement.

TECHNICAL SKILLS

Industrial Controls: Siemens S7-1200 and LOGO!, PLC sequencing, contactors, VFDs, three-phase diagnostics, sensors, actuators, motor/drive troubleshooting, operator controls, basic HMI/SCADA concepts.

Maintenance & Equipment: CNC routers, material handling and conveyors, automated production equipment, compressors, dust collection, electric/mechanical troubleshooting, planned maintenance, vendor/contractor coordination, root-cause investigation.

Software & Integration: Python, Flask, Flask-SocketIO, Pandas, OpenPyXL, JavaScript, Excel reporting, barcode systems, industrial PCs/networking, G-code processing, CAD/CAM utilities, Git/GitHub.

Engineering Methods: Control wiring, commissioning and functional testing, technical documentation, process standardization, cross-training, production KPI analysis, CAD/CAM, prototyping and troubleshooting.

EDUCATION & DEVELOPMENT

Bachelor of Technology, Electronics & Communication Engineering **Punjabi University, India**

Automated Controls Installation & Maintenance – Advanced Certificate **BCIT**
In progress; coursework in PLCs, HMI, drives/motion, instrumentation and industrial control systems.

International Business Management **St. Lawrence College, Canada**

Lean Six Sigma White Belt **Completed Sep 2026**

PROJECT PORTFOLIO

bajwa95.github.io Selected case studies with photographs and technical implementation details in manufacturing systems, PLC retrofits, CNC troubleshooting, and equipment reliability.